

Regulations: 2022

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Reg No:

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BANNARI AMMAN INSTITUTE OF TECHNOLOGY
(An Autonomous Institution Affiliated to Anna University, Chennai)
SATHYAMANGALAM - 638 401

Periodical Test - II - October 2025

S7 Semester

Answer Key

22AI702 & AI FOR ROBOTICS

Degree & Branch: B.Tech. & AD

Maximum: 50 Marks

Time: 1:30 hrs

Instructions:

- Students should not mark/write anything on the Question Paper other than the register number.
- Section A of the Question Paper contains questions for 15 Marks. Sections B and C contain questions for 30 Marks each.
- Students can attempt answering any two out of three subsections in each section. The maximum mark is limited to 10 in section A and 20 in section B & C.

Questions

Q. No.

Section - A

COURSE OUTCOME 3

MAXIMUM: 10 MARKS

A1

(i)

d) Landmarks for localization

(ii)

Q1 – Identify (2 Marks):
Wheel Encoder Drift → Errors accumulate due to wheel slippage or uneven terrain. (1 Mark)
Sensor Noise → Camera/LiDAR may give inaccurate data in poor lighting or cluttered environment. (1 Mark)
Q2 – Analyze (2 Marks):
Localization → Small position errors accumulate over time, leading to inaccurate robot location. (1 Mark)
Path Planning → Incorrect position estimation can cause inefficient or unsafe routes. (1 Mark)

A2

(i)

a) Localization becomes less accurate due to loss of stable reference points.

(ii)

a) Prediction (1M):
False – Wheel encoder errors contribute to localization drift, but combining with LiDAR (sensor fusion) can reduce long-term inaccuracies.
b) Evaluation (3M):
Errors in wheel encoders accumulate due to slippage, uneven surfaces, and mechanical inaccuracies.
LiDAR errors occur from noise or occlusions in the environment.
Accumulated errors affect the robot's position estimation, leading to deviations from planned paths and possible collisions.
Sensor fusion algorithms, such as Kalman filters, help correct these errors and maintain accurate navigation.

A3

(i)

a) Path guidance features

	(ii)	1. Identify (1 Mark):Correct statements: a and c 2. Support (2 Marks): Statement a: Feature extraction converts raw sensor data into meaningful features like shelf corners or pallet edges, enabling the robot to recognize key points in the environment for navigation. Statement c: Extracted features allow the robot to localize itself, detect moving forklifts, and plan safe paths, which ensures accurate and collision-free movement. 3. Analyze (1 Mark): Skipping feature extraction would reduce the robot's perception of the environment, leading to poor localization, higher chances of collisions, and inefficient or unsafe navigation paths.	
Section - B		COURSE OUTCOME 4	MAXIMUM: 20 MARKS
B1	(i)	b) Multi-sensor redundancy	
	(ii)	b) Particle filter	
	(iii)	a → e → d → b → f → c a. The Drone assumes a starting position with some confidence but still considers nearby alternatives. (0.5 Mark) e. Small motion errors (motor noise, air disturbance) make its exact position less certain. (0.5 Mark) d. The probability spreads over multiple possible positions, reflecting increased uncertainty.(0.5 Mark) b. The drone gathers data from cameras, sonar, or LIDAR about walls, doors, or obstacles. (0.5 Mark) f. Sensor data eliminates unlikely positions and strengthens the probability of correct ones. (0.5 Mark) c. The refined distribution gives the most likely current position, ready for the next cycle. (0.5 Mark)	
	(iv)	1.Noise definition + effect → Sensor readings fluctuate due to reflections, causing inaccurate distance estimates. (2 marks) 2. Aliasing definition + effect → Identical corridors lead to location confusion, since sensor data cannot distinguish them. (2 marks) 3. Overall impact → Both challenges increase localization errors, making reliable navigation difficult without sensor fusion or probabilistic filtering. (1 mark)	
B2	(i)	c) Semantic map	
	(ii)	a) GPS is not available underground	
	(iii)	1,2,4,6 1. Maps allow the robot to plan patrol routes efficiently. (1 mark) 2. Maps help avoid collisions with walls, escalators, or people. (1 mark) 4. Map representation enables coordination with other security robots. (1 mark) 6. Maps help the robot quickly locate areas that need urgent attention. (1 mark)	
	(iv)	The robot is likely to use an Occupancy Grid Map (OGM) representation. (1 Marks) Point 1: OGM divides the environment into grid cells marked as free, occupied, or unknown, allowing precise navigation even in narrow corridors and crowded spaces. (1 Mark) Point 2: The grid structure supports real-time updates, enabling the robot to handle dynamic obstacles like moving people or temporary stalls. (1 Mark) Point 3: OGM is well-suited for large indoor spaces without GPS, ensuring the robot can maintain accurate path planning across multiple levels of the mall. (1 Mark)	

B3	(i)	b) It builds and updates maps while localizing
	(ii)	b) Sensor fusion for localization
	(iii)	1.Ensures accurate path following by tracking the robot's real-time position. (1 mark) 2.Provides dynamic obstacle avoidance, enabling safe rerouting when objects block the path. (1 mark) 3.Improves delivery efficiency and reliability by minimizing delays and wrong turns. (1 mark)
	(iv)	1 → c 2 → d 3 → a 4 → b 5 → e

Section - C	COURSE OUTCOME 5	MAXIMUM: 20 MARKS
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C1	(i)	b) Tabu search by storing explored paths
	(ii)	a) SLAM
	(iii)	C → E (1 m) → B → F (1 m) → A → D (1 m) C. The robot senses its environment in real-time using LIDAR, cameras, or other sensors. E. The robot identifies dynamic obstacles based on proximity and movement patterns. B. AI models track the direction and speed of moving objects to predict their motion. F. It uses the Dynamic Window Approach (DWA) to adjust its path in real-time. A. The robot reduces speed, keeps a safe distance, or waits when near humans to ensure safety. D. When the local path is blocked, the robot updates its global route to continue its task.
	(iv)	a.  The robot reaches static equilibrium at the goal and comes to a natural stop without external disturbance. (1 mark) b. The robot is in equilibrium — it experiences no motion-driving forces. (1 mark)
C2	(i)	c) Distributed system with shared environment models
	(ii)	a) Use sensor fusion techniques to combine readings intelligently

(iii)	
(iv)	Deliberative navigation relies on a predefined global map, assuming the environment is static. (1 mark) Unexpected obstacles, such as newly placed shelves or temporary blockages, cannot be accounted for in advance. (1 mark) The robot is unable to update its path in real time, causing it to halt or become inefficient. (1 mark) This approach works well in predictable and structured environments but struggles in dynamic settings. (1 mark) To make it more adaptable, a reactive layer can be added to detect and avoid unforeseen obstacles, forming a hybrid navigation system. (1 mark)
C3	(i) d) Recompute the path based on updated map
	(ii) c) Take an alternative route using stairs or another elevator
	(iii) Type: Reactive Motion Planning (1 mark) Function: Computes feasible and safe velocity commands for the robot over a short time horizon based on current sensor inputs. (1 mark)
	(iv) 1. Represent the Environment as a Grid The warehouse is modeled as a 2D grid, where: Each cell = one unit of space 0 = free cell 1 = obstacle (e.g., racks or walls) Start position: (0, 0) Goal position: (5, 5) (2 mark) 2. Define the Cost Functions A* uses the function: $f(n)=g(n)+h(n)$ (1 mark) 3. Explore Neighbors A* starts at (0,0), adds it to the open list. It explores neighboring cells (up, down, left, right—diagonal if allowed). (1 mark) 4. Choose the Best Node The node with the lowest $f(n)$ value is chosen for expansion. The process repeats, updating g and f for all reachable nodes. Obstacles (1) are skipped and not added to the open list. (1 mark) 5. Reach the Goal When the robot reaches (5,5), A* traces back from the goal to the start using parent pointers. The resulting path is the optimal shortest path, avoiding all obstacles. (1 mark)

*** End of Question Paper ***